Title:

Dynamic Relationship Between Gold and Silver Prices – daily data

Course:

Statistical Methods for Time Series

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1.0 Introduction

1.1 Background of study

Gold and silver have been recognized as prominent assets in the financial markets because of their use as store-of-value assets, investing tools, as well as hedges against inflation and uncertainty. Gold has been noted as a safe-haven asset in many studies, especially during financial stresses, while silver has been seen as volatile due to its intrinsic property as a precious metal, as well as its usage as an industrial material (Baur & Lucey, 2010).

Empirical observations reveal that gold and silver prices tend to move together, thereby implying a close linkage between both markets. Though their prices may differ for a short while owing to certain market-related shocks, investor sentiments, or demand variations, both appear to share a similar trend over a long period of time. This trend generates a possibility regarding a long-run equilibrium relationship between gold and silver prices, where deviations between both prices get eradicated over a period of time.

From the time series econometric point of view, it is noted that several works stress that the time series of the prices of commodities are normally non-stationary. (Engle & Granger, 1987) identify the idea of cointegration. They show that it is possible that a relationship exists among the non-stationary time series towards a stable equilibrium. This model comes into application for the analysis of financial and physical markets for precious metals. However, if cointegration does exist, the changes that occur for a shorter time period could be obtained from the ECM model.

With these findings as the background, this study uses recent techniques of time series analysis to investigate how gold and silver prices are related dynamically. By doing so, this study can also help to supplement the existing literature with regard to precious metals while providing insights that will be relevant for investors and risk management.

1.2 Objectives of study

The main objectives of this study are as follows:

1. To examine whether gold and silver prices share a long-run equilibrium relationship(cointegration).
2. To analyse the short-run dynamic interactions between gold and silver prices and how do short-term shocks in one affect the other.
3. To identify which metal adjusts to deviations from the long-run equilibrium in the short run.

2.0 Literature Review

Many studies examine the relationship between gold and silver prices because of their importance in global financial and commodity markets. Numerous studies look at whether the values of gold and silver move in parallel over time and how their interactions alter under different market conditions because both metals are traded globally and frequently react to similar macroeconomic factors. But because silver has a larger industrial component and gold is primarily an investment asset, their demand patterns are different, indicating that their price dynamics may not be the same, especially in the short term.

The prices of gold and silver are examined in several empirical studies within a larger framework of financial and commodity markets. The dynamic relationships between oil prices, exchange rates, and the prices of precious metals, such as gold and silver, are studied by Sari, Hammoudeh, and Soytas (2010). They discover through time-series analysis that common external shocks like changes in oil prices and exchange rates have an impact on the prices of gold and silver. Their findings do, however, also demonstrate that the magnitude and durability of these effects vary among metals. This implies that although gold and silver frequently react to similar economic forces, their short-term behaviour varies due to the strength of their reactions.

Other research focusses more intently on the relationships between volatility in the markets for precious metals. The volatility spillovers between precious metals and associated financial markets are examined by Hammoudeh, Yuan, McAleer, and Thompson (2010). According to their findings, the silver market is frequently affected by increases in the volatility of the gold market, especially during times of increased uncertainty. Stronger volatility responses are observed in silver, which may be due to its greater sensitivity to industrial demand conditions and speculative activity. These results show that even when the prices of gold and silver fluctuate in tandem over time, their risk dynamics can differ significantly.

Studies on commodity prices usually emphasise the potential for long-term changes in the correlations between commodity prices. Pindyck (1999) demonstrates how long-term economic factors affect commodity prices without concentrating only on precious metals, even though they can also change as a result of structural shifts, financial shocks,

and shifts in demand worldwide. This suggests that the purported connections between the gold and silver markets may not remain true over time, particularly if the sample includes notable economic events or disasters.

A recurrent element in this research is the fact that while gold and silver prices are affected by similar economic causes, they behave differently. Short-term price dynamics and volatility responses can vary significantly, even though long-term co-movement is frequently seen. Additionally, several authors note that the time period and market conditions under consideration have a significant impact on empirical results. Shorter sample sizes may make it difficult for studies to capture how relationships change under various economic regimes.

Although their short-term dynamics and volatility behaviour vary depending on market conditions, the empirical literature generally indicates that gold and silver prices are closely related over longer time horizons. To better understand how the relationship between gold and silver prices changes over time, these findings encourage additional empirical research using longer datasets that span several economic cycles.

3.0 Data and Methodology

3.1 Hypothesis development

Informed by the preceding literature review, the following hypotheses are empirically tested:

Hypothesis 1: Gold and silver prices are cointegrated and share a common long-run equilibrium relationship.

Hypothesis 2: Silver prices respond to deviations from the long-run equilibrium, while Gold prices may does not exhibit significant error correction behavior.

Hypothesis 3: Gold and Silver prices exhibit Granger causality(short-run) relationship.

3.2 Data Description

The dataset that was used in our research was sourced from *Yahoo Finance*[1]. Data of Gold and Silver are daily and from January 2007 to December 2025. We conduct all tests and regressions in R.

3.3 Data transformation

To prepare the gold and silver price series for time series analysis, several data transformation steps were applied. The analysis focuses on daily closing prices, which are commonly used in financial studies due to their ability to reflect the final market valuation of each trading day.

First, the series were examined for missing observations. Since time series models require continuous data, missing values were replaced using linear interpolation. This method estimates missing values based on neighbouring observations and helps maintain the overall structure of the series without introducing abrupt distortions.

For visualization purposes, the gold and silver price series were also standardized by subtracting their respective means and dividing by their standard deviations. This transformation allows both series to be plotted on the same scale and facilitates comparison of their relative movements. It is important to note that this standardization is used only for descriptive analysis and not for model estimation.

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[1] Yahoo Finance - Stock Market Live, Quotes, Business & Finance News

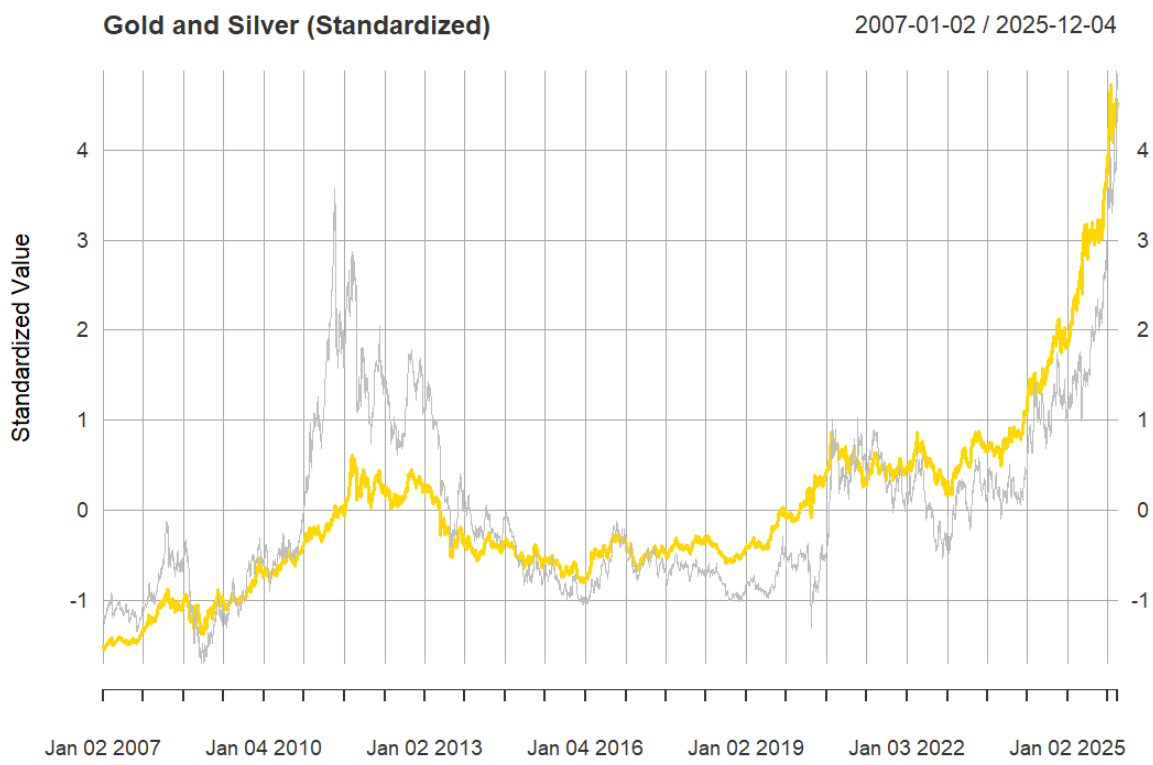

Figure 1: Time series of gold and silver closing price

In Figure 1, we standardized the price series by subtracting the mean and dividing by the standard deviation. This tells us how far the current price is from its own long-term average, measured in standard deviations, which helps us see co-movement and relative strength. By doing this, prices with different units are brought onto the same scale, allowing us to focus on their movements instead of their price levels. This standardization is used only for this figure. In the rest of the analysis, we work with the original price levels.

We can observe from this figure, between gold and silver over the sample period, that their prices tend to move together over time. However, at the same time, their relative strengths varied significantly across different subperiods. For instance, from the late 2000s to the early 2010s, silver prices showed a more pronounced positive deviation from their historical mean, whereas after 2020, gold demonstrated relatively stronger performance. These observations raise the research question, whether the long-run relationship between gold and silver prices remain stable over time.

Finally, because financial price series often exhibit exponential growth and changing variance over time, a logarithmic transformation was applied to both gold and

silver prices, making the series more suitable for stationarity testing and subsequent econometric modelling.

3.4 Model

In contrast to cross-sectional data, which is typically assumed to be independent and identically distributed (i.i.d.), time series data exhibits temporal dependence and may display non-stationarity. Consequently, the distribution of observations in this study cannot be assumed to remain constant over time, as both the mean and variance may vary across different time periods. Cross-sectional OLS regressions based on such data may fail to produce reliable or universally valid population estimates. Under non-stationary conditions, even uncorrelated series may exhibit correlations. Non-stationarity typically stems from trends, seasonality, and exogenous shocks (such as the COVID-19 pandemic or economic crises).

Cointegration provides an appropriate framework for addressing this issue, as it allows for short-run deviations between non-stationary series while explicitly modeling their potential long-run equilibrium relationship. Specifically, we are interested in whether the gold and silver prices share a common long-term trajectory, implying a stable long-term equilibrium despite temporary divergences.

3.4.1 Seasonality

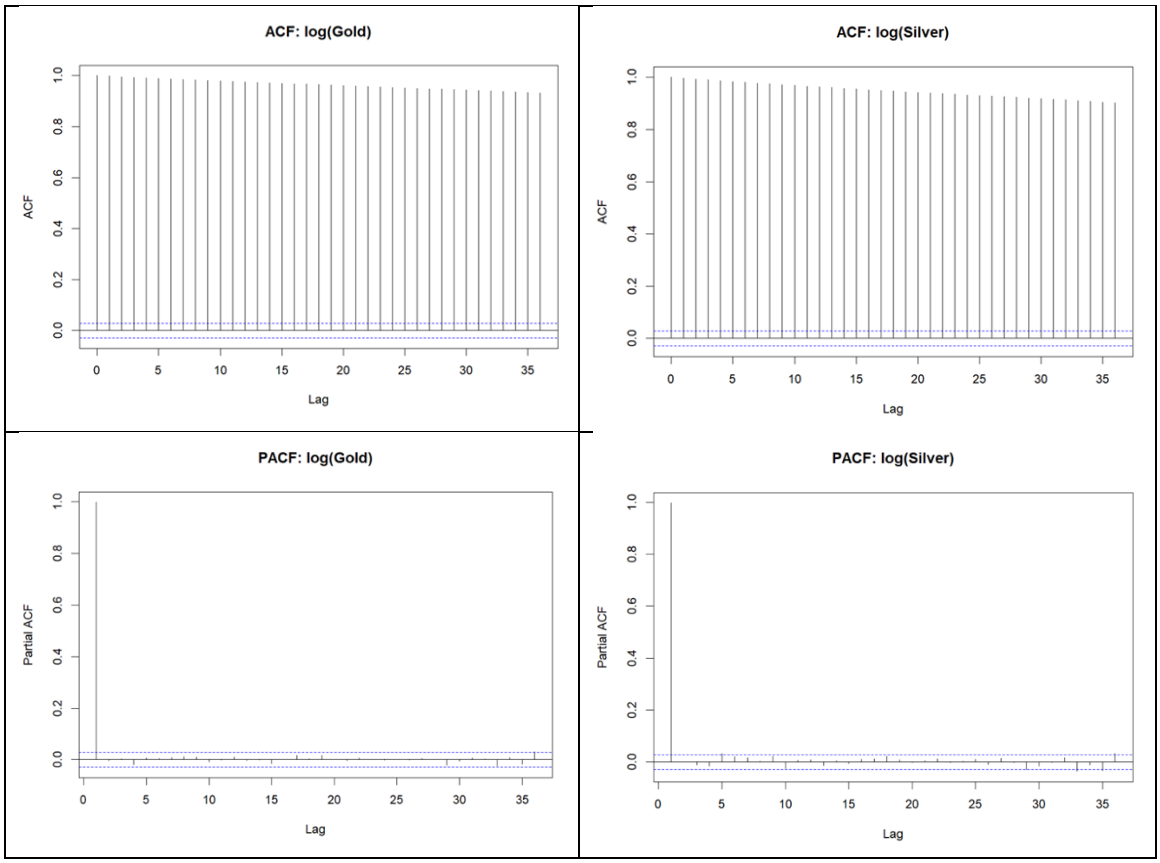

Figure 2: Autocorrelation functions of log-transformed gold and silver prices

As shown in Figure 2, display the autocorrelation functions of log-transformed gold and silver price series. No pronounced seasonal patterns were detected. This finding aligns with market efficiency theory, which is any persistent and predictable seasonal price pattern would likely be exploited by arbitrage activities, thereby eliminating such patterns in equilibrium.

3.4.2 Stationarity

As shown in the autocorrelation function and partial autocorrelation function plots present before, the series exhibits strong persistence, with the autocorrelation coefficients decaying extremely slowly at lag values. This pattern indicates that the sequence is likely non-stationary.

Furthermore, the combination of gradually decaying autocorrelation functions and a partial autocorrelation function that truncates after the first few lags is consistent with an AR model.

To formally assess the stationarity of this series, we will conduct stationarity tests using the ADF test and the KPSS test.

Table 1: Stationarity Test Results for the log(Gold) Series

Method	Result
KPSS	Reject H0, provide evidence that the series is non-stationary.
ADF	Fail to Reject H0, indicating that the series is non-stationary.

Table 2: Stationarity Test Results for the log(Silver) Series

Method	Result
KPSS	Reject H0, provide evidence that the series is non-stationary.
ADF	Fail to Reject H0, indicating that the series is non-stationary.

Table 1and Table 2 reports the stationarity test results for the log-transformed gold and silver price series. The KPSS test rejects the null hypothesis of stationarity, while the ADF test fails to reject the null hypothesis of a unit root. Taken together, these results provide strong evidence that both log (Gold) and log (Silver) series are non-stationary. Therefore, we applied first-order difference to both our time series and do the ADF test and the KPSS test again.

Table 3: Stationarity Test Results for first-order differenced log(Gold) series	
Method	Result
KPSS	Fail to Reject H0, indicating that the series is stationary.
ADF	Reject H0, indicating that the series is stationary

Table 4: Stationarity Test Results for first-order differenced log (Silver) Series	
Method	Result
KPSS	Fail to Reject H0, indicating that the series is stationary.
ADF	Reject H0, indicating that the series is stationary

Tables 3 and 4 report the stationarity test results for the first-order differenced log-transformed gold and silver price series, respectively. For both series, the KPSS test fails to reject the null hypothesis of stationarity, while the ADF test rejects the null hypothesis of a unit root.

Overall, the results from both tests consistently suggest that the log(Gold) and log(Silver) price series are first-order integrated, $I(1)$.

3.4.3 Cointegration

Next step is to determine the most appropriate econometric framework for modelling the relationship between gold and silver prices.

Given that both log(Gold) and log(Silver) are non-stationary in levels but stationary after first differencing, the series are classified as I(1). Under such conditions, two modelling approaches are possible. If the variables are not cointegrated, a Vector Autoregressive (VAR) model in first differences is appropriate, as it avoids spurious regression problems. However, if the variables are cointegrated, an Error-Correction Model (ECM) should be employed to preserve the long-run equilibrium relationship while modelling short-run adjustments.

In this study, cointegration is examined using the Engle–Granger methodology, which If a X and a Y time series are not stationary by themselves, meaning they have a unit root, however if I build a simple OLS model on them and the residuals of the model are stationary, then the X and Y time series are cointegrated.

Until now, we get the log(Gold) and log(Silver) price series are first-order integrated, $I(1)$. We are going to test if there are long-run equilibrium relationship exists between them. Therefore, we did a smooth OLS regression for the two-time series.

The choice of dependent variable in the Engle–Granger regression does not affect the existence of cointegration, we are more focus on the residuals, here we do regress log(Gold) on log(Silver). Then we do the ADF test and the KPSS test on their residuals.

Table 5: Stationarity Test Result on Residuals				
Test	Test Statistic	1% CV	5% CV	10% CV
KPSS (EG residuals)	0.1362	0.739	0.463	0.347
ADF (EG residuals)	-2.2765	-2.58	-1.95	-1.62

Therefore, the formular are following:

$$log(Gold_t) = 4.591 + 0.886log(Silver_t) + u_t$$

Note that the reported critical values are not true Engle–Granger critical values. Hence, we focus primarily on the relative magnitude of the test statistics rather than formal significance levels. Overall, the ADF rejects the null hypothesis of a unit root at the 5% significance level, but fails to reject it at 1% significance level, while the KPSS test fails to reject the null hypothesis of stationarity with any common significance level. These results together provide evidence that the residual series is stationary. Therefore, it is acceptable to say that the two series, which is log(Gold) and log(Silver) are cointegrated. Which means that they have a common long-term trajectory, from which deviations from the long-run relationship are temporary and that the series tend to move together over time. Moreover, the estimated coefficient suggests that a 1% increase in silver prices is associated with a 0.88% increase in gold prices in the long run.

3.4.4 Error-Correction Model (ECM)

Up to this point, we have proved that our time series are cointegrated, that means they have a common long-term trajectory. The next step is to what extent the time series stick to this long-term path from one period to the next, in other words, after a deviation, will they return to equilibrium? How quickly will they recover?

The VECM can be interpreted as a VAR representation of first-order differenced variables augmented with an error-correction term that captures deviations from the long-run equilibrium, in this case, the first-order differenced log(Gold) and log(Silver), which are both stationary, and with error correction variables. This error correction variable is the first lag of the error term that was found to be stationary in the Engle-Granger test. The error-correction term measures the extent to which the system deviates from its long-run equilibrium in the previous period. Its coefficient reflects the speed at which short-run deviations are corrected.

By combining short-run dynamics with long-run equilibrium behaviour, the ECM provides a comprehensive representation of the relationship between gold and silver prices. This makes it particularly suitable for financial time series that exhibit both persistent trends and short-term fluctuations.

Table 6: Lag Length Selection based on First-Order differenced variables

Information Criterion	Selected Lag
AIC	3
HQ	2
SC (BIC)	1
FPE	3

Lag length selection was conducted using the VARSelect function based on the first-order differenced variables. We are now strictly penalizing extra explanatory variables, therefore, although AIC and FPE suggest a higher lag order, according to the BIC which is SC, the optimal lag length is one. Therefore, we decided the lag order 1 are subsequently adopted in the error correction model specification.

Table 7: Error-Correction Model (ECM) Estimation Results for Gold and Silver Prices				
Dependent Variable	ECT	Δlog(Gold)_{t−1}	Δlog(Silver)_{t−1}	Intercept
Δlog(Gold)_t	0.0006 (0.0003).	0.0203 (0.0237)	−0.0171 (0.0129)	0.0004(0.0002)*
Δlog(Silver)_t	0.0012(0.0006)*	0.0734 (0.0436).	−0.0490 (0.0237)*	0.0002 (0.0003)
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1				

As shown in the table, the error correction term (ECT) in the Δlog(Silver) equation is statistically significant at the 5% level, whereas in the Δlog(Gold) equation it is only marginally significant at the 10% level and insignificant at the 5% level, indicating that silver prices respond to deviations from the long-run equilibrium, while gold prices show weaker responsiveness. The small magnitude of the ECT coefficient suggests that the short-run response of silver prices to long-run deviations is relatively weak, which means the adjustment process toward the long-run equilibrium appears to be slow.

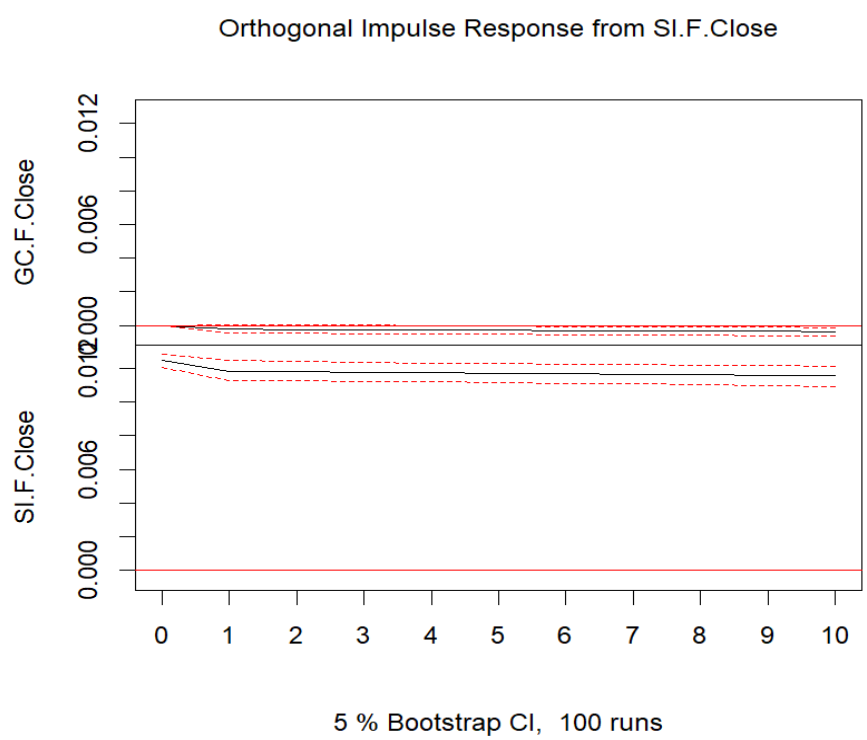

Figure 3. Orthogonal Impulse Response Functions (IRFs) of Gold Prices Following a One-Standard-Deviation Shock in Silver Prices

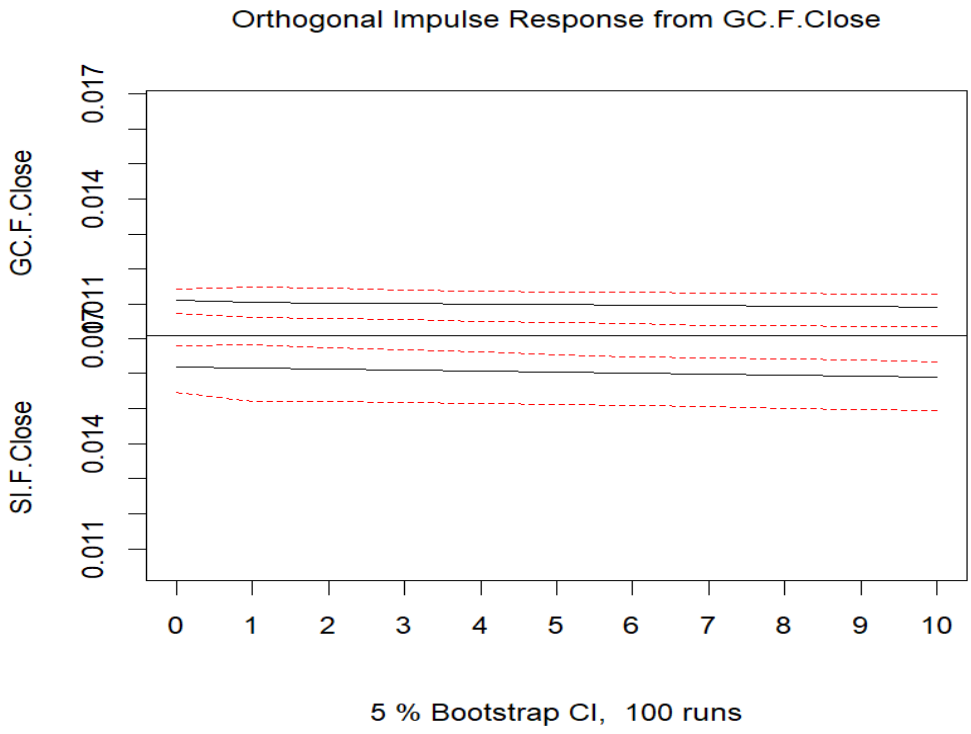

Figure 4. Orthogonal Impulse Response Functions (IRFs) of Silver Prices Following a One-Standard-Deviation Shock in Gold Prices

In addition, we see that the Δlog(Silver) has a classic AR(1) component, which means that, the current value significantly decreases if the price increased yesterday, implying that short-run changes in silver prices are partially driven by their own past one period value. By contrast, none of the explanatory variables in the Δlog(Gold) equation are statistically significant. Although the lagged change in gold prices in the silver equation is marginal significance at the 10% level, however, it is insignificant at the 5% level.

Taken together, these results provide no strong evidence of short-run Granger causality between gold and silver prices. Which means, in the short term, the two time series do not have a significant influence on each other.

Hence, the relationship between the two series is mainly characterized by a long-run equilibrium, with limited short-run interactions. In specifically, the adjustment toward the long-run equilibrium is primarily driven by changes in silver prices, while gold prices appear weakly exogenous with respect to the long-run relationship, as deviations from the long-run equilibrium are primarily reflected in adjustments in silver prices.

Finally, we test the stationarity of residuals of each equation, make sure they are white noise.

Table 8: Residual Diagnostic Test Results for the Error-Correction Model		
Equation (Residuals)	p-value	Conclusion (5% level)
Δlog(Gold)	0.0453	Reject H_0
Δlog(Silver)	0.0587	Fail to reject H_0

As shown in Table 8, the residuals of the Δlog(Silver) equation behave like white noise, as the null hypothesis of no serial correlation cannot be rejected at the 5% significance level.

By contrast, the Breusch–Godfrey test p-value for the Δlog(Gold) residual is 0.0453, which is below the 5% significance level. Therefore, the null hypothesis of no serial correlation is rejected for the Δlog(Gold) equation, indicating that its residuals cannot be considered white noise.

Importantly, the p-value is very close to 0.05, given that the lag order is selected based on the BIC, and the detected correlation is weak and borderline, therefore, the baseline VECM specification is retained for subsequent analysis, and is our final model.

3.5 Final Model Interpretation

$$log(Gold_t) = 4.591 + 0.886log(Silver_t) + u_t$$

This study analyzes the relationship between gold and silver prices using daily data from 2007 to 2025. After transforming the prices into logarithms, several stationarity tests show that both gold and silver prices are non-stationary in levels but become stationary after first differencing. This indicates that both series are integrated of order one, which allows for cointegration analysis.

The cointegration results suggest that gold and silver prices share a stable long-run relationship. Although the two prices may diverge in the short run, they do not move independently over time. Instead, deviations from their long-term equilibrium tend to be corrected, which is reasonable given that both metals are affected by similar economic and market factors.

A Vector Error Correction Model (VECM) is used to capture both short-run dynamics and long-run adjustment. The results indicate that the relationship between the two series is mainly characterized by a long-run equilibrium, with limited short-run interactions, which means that short-term price changes respond to past deviations from the long-run relationship, and in the short term, the two time series do not have a significant influence on each other. Overall, the findings confirm a strong long-term link between gold and silver prices with short-term fluctuations.

4.0 Conclusion

This paper examines the relationship between gold and silver prices using time-series methods over the period from 2007 to 2025. The results show that both prices are non-stationary in levels but stationary in first differences.

Cointegration analysis provides evidence of a stable long-run relationship between gold and silver prices, suggesting that the two markets are closely connected, while short-run interactions between the two markets are limited.

The Vector Error Correction Model confirms that deviations from the long-run equilibrium are mainly corrected through adjustments in silver prices, whereas gold prices exhibit weak responsiveness.

Furthermore, Impulse Response Functions and Granger causality tests provide no strong evidence of short-run shock transmission between the two series.

These findings imply that the gold price and silver price relationship is primarily demonstrates by long-term co-movement rather than short-term dynamic dependence.

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